Beginnings in Algebraic Geometry Notes for CH 9 - 12

SFSU FALL 2026

Contents

1 CH 9 Projective Varieties **2**
- 1.1 9.1 Projective Space . . . 2
- 1.2 9.2 Projective Vanishing Operator . . . 3
- 1.3 9.3 Projective $\mathcal{I}$-operator . . . 4
- 1.4 9.4 Affine Restrictions . . . 5
- 1.5 9.5 Projective Closure . . . 6
- 1.6 9.6 Project Nullstelensatz . . . 8

2 CH 10 Maps of Projective Varieties **10**
- 2.1 10.1 Regular Maps of Projective Varieties . . . 10
- 2.2 10.2 Projective Equivalences . . . 11
- 2.3 10.3 Veronese Maps . . . 12
- 2.4 10.4 Segre Maps . . . 15

1 CH 9 Projective Varieties

1.1 9.1 Projective Space

Definition 1.1 (Projective Space). *Let $n \in \mathbb{N}$ a projective space over K denoted $\mathbb{P}^n_k$ is defined as*

$$\frac{K^{n+1}/\{(0,\dots,0)\}}{\sim}$$

where $(a_0,\dots,a_n) \sim (b_1,\dots,b_n)$ iff there exists a $\lambda \in K/\{0\}$ such that

$$(a_0,\dots,a_n) = (\lambda b_0,\dots,\lambda b_n).$$

An equivalence class is denoted $[a_0 : \cdots : a_n]$.

Remark 1.1. Note here that we aren't really putting any structural requirements on K. My previous encounters with projective spaces almost always used a field and the notation of K is consistent with the textbooks use of fields. With affine spaces we noted that the Nullstellensatz was only consistent when K was an algebraically closed field. I imagine we will soon face a similar restriction. *Does K need to be a field?*

Projective spaces are particularly interesting because there are many related ways of presenting them. The first thing we'll do as a responsible mathematicitians is go through a couple of examples.

Ex Let $n = 0$, that is $\mathbb{P}^0_K$ for some K. This means we are looking at equivalence classes of $\frac{K/\{0\}}{\sim}$. Note that for any $a \in K/\{0\}\}$ we can always take $\lambda = \frac{1}{a}$ therefore all of $\mathbb{P}^0 = \{[1]\}$.

Ex Take $n = 1$, for any $[a_0 : a_1] \in \mathbb{P}^1$ observe that by taking $\lambda = \frac{1}{a_0}$ then every equivalence class where $a_0 \neq 0$ is simply $[1 : \frac{a_1}{a_0}]$. On the other hand when the first term is zero, $[0 : a_1]$ we may always choose $\lambda = \frac{1}{a_1}$ so that it may be scaled to $[0 : 1]$. This means that $\mathbb{P}^1$ can instead be expressed as a disjoint union

$$\mathbb{P}^1 = \{[1 : a] \mid a \in K\} \sqcup \{[0 : 1]\}.$$

Observe that the left term is in bijection with $\mathbb{A}^1$

Ex Let $n = 2$. Similar to $\mathbb{P}^1$, equivalence classes $[a_0 : a_1 : a_2] \in \mathbb{P}^2$ can always be rescaled to $[1 : b_1 : b_2]$ when $a_0 \neq 0$. If the first term is zero then again every term can be scaled to $[0 : b_1 : b_2]$. Then,

$$\mathbb{P}^2 = \{[1 : b_1 : b_2] \mid b_1, b_2 \in K\} \sqcup \{[0 : b_1 : b_2] \mid b_1, b_2 \in K\}$$

Interestingly the by dropping the first term; the left portion is in bijection with $\mathbb{A}^2$ while the right term is in bijection with $\mathbb{P}^1$ which gives the decomposition

$$\mathbb{P}^2 = \mathbb{A}^2 \sqcup \mathbb{P}^1.$$

Theorem 1.2. *For $n \geq 1$ that there exists a natural bijection $\mathbb{P}^n = \mathbb{A}^n \sqcup \mathbb{P}^{n-1}$*

Proof. The proof follows the structure provided in problem 9.1.3 and consists of three portions. First define

$$U = \{[a_0 : \cdots : a_n] \in \mathbb{P}^n \mid a_0,\dots,a_n \in K, a_0 \neq 0\} \quad \text{and} \quad V = \{[a_0 : \cdots : a_n] \in \mathbb{P}^n \mid a_0,\dots,a_n \in K, a_0 = 0\}$$

We begin by showing that $\mathbb{P}^n = U \sqcup V$. The containment of $\mathbb{P}^n$ in $U \cup V$ and vice verse is automtic given the definition of U and V. It remains to show that $U \cup V$ can be upgraded to $U \sqcup V$. Suppose towards a contradiction that $U \cap V \neq \emptyset$ then there exists $[b_0 : \cdots : b_n] \in U \cap V$. Note that for any $[a_1 : \cdots : a_n] \in U$, because a_0 is non-zero, we may always take $\lambda = \frac{1}{a_0}$ so that every equivalence class contained in U is of the form $[1 : c_1 : \cdots : c_n]$ for some $c_1,\dots,c_n \in K$. The assumption that $U \cap V$ is non-empty implies that b_0 is both zero and one. Hence $U \cap V = \emptyset$ and $\mathbb{P}^n = U \sqcup V$. The isomorphism $V \simeq \mathbb{P}^{n-1}$ and $U \simeq \mathbb{A}^n$ follow immediately by simply dropping the leading zero or one. ℵ

There are a few more ways that projective spaces are represnted. The most well known is that $\mathbb{P}^n$ is isomorphic to the set of points through the origin of $\mathbb{A}^{n+1}$

1.2 9.2 Projective Vanishing Operator

We say that a polynomial $f \in K[x_0, \ldots, x_n]$ only vanishes on a point a if it vanishes on every representative of a in the desired projective space.

Non-Example: Consider the polynomnial $f(x, y) = x^2 - y$ which vanishes on $[2, 4]$ but it doesnt vanish on $[4, 8]$

Example: The polynomial $f(x, y) = x^2 - y^2$ on the other hand vanishes on $[2, 2]$ and every possible representative of the solution

We end up with a problem: How do we check that every representative vanishes? Instead of checking we simply restrict ourselves to the case of homogenous polynomials. A polyomial is considered homogenous of degree d if every non-zero term of f has degree d. For example $f(x, y, z) = 3x^2 + 2yz + 3zx$ is a polynomial homogenous of degree 2 because every non-zero term has degree 2.

Proposition 1.1. *Let $f \in K[x_0, \ldots, x_n]$ be a homogenous polynomial of degree d and $[a_0 : \cdots : a_n] = [b_0 : \cdots : b_n]$ in $\mathbb{P}^n$ then $f(a_0, \ldots, a_n) = 0 \iff f(b_0, \ldots, b_n) = 0$.*

Proof. If f is a homogenous polynomial of degree d, then each non-zero term of f takes the form $cx_0^{\alpha_0} \cdots x_n^{\alpha_n}$ for some non-zero $c \in K$, $\alpha_0, \ldots, \alpha_n \in \mathbb{N}$ such that $\sigma\alpha_i = d$. Evaluating f at a representative of $[a_0 : \cdots : a_n]$ means each non-zero term takes the form

$$c\lambda^{\sum_i \alpha_i} a_0 \cdots a_n = c\lambda^d a_0 \cdots a_n.$$

Then λ^d can be factored out of every non-zero term of f evaluating at so that $f(\lambda a_0, \ldots, \lambda a_n) = \lambda^d f(a_0, \ldots, a_n)$. If $f(a_0, \ldots, a_n) = 0$ and, by assumption, there exists a $\lambda \in K$ such that

$$(\lambda a_0, \ldots, \lambda a_n) = (b_1, \ldots, b_n)$$

with the desired implication following direcrtly. Following identical logic the reverse implication completes the desired if and only if. ℵ

Of course in true algebraist sense, the fact that we can determine that a homegenous polynomial vanishes on all representatives as long as it vanishes on one, we restrict ourselves to looking only at homogenous polynomials. Yet we would like to still work with non-homogenous as a strict homogeneity condition is too restrictive. Lucky for us we can considere homogenous components of a polynomial. For example the polynomial $f(x, y) = x^2 + xy + x$ has a homogenous component of degree 2 and degree 1 denoted

$$f_2(x, y) = x^2 + xy \quad \text{and} \quad f_1(x, y) = x$$

. The next proposition follows quite naturally

Proposition 1.2. *Let $f \in K[x_0, \ldots, x_n]$ be a polynomial and $a \in \mathbb{P}^n$. Then f vanishes at a if and only if each homogenous component of f vanishes at a.*

Proof. Suppose that each homogenous component f_k of f vanishes on a for all $k \leq d$ where d is the degree of polynomial f. As

$$f = \sum_{k=1}^{d} f_k$$

and $f_k(a) = 0$ for every representative of a, then f vanishes at every representative of a. Now suppose that f vanishes on every representative of a. Then for any $\lambda \in K \setminus \{0\}$, by assumption it follows

$$0 = f(\lambda a_0, \ldots, \lambda a_n) \tag{1}$$

$$= \sum_{k=1}^{d} \lambda^d f_k(a_0, \ldots, a_n). \tag{2}$$

Since (2) vanishes of inifnitely many λ we may view (2) as a univariate polynomial

$$f(x) = \sum_{k=1}^{d} x^d f(a_0, \ldots, a_n) = 0$$

vanishes on infinitely many values of K. This implies that $f(x)$ must be the zero polynomial then, $f_k(a_0, \ldots, a_n) = 0$ for all k. Since f_k is homogenous then each homogenous component of f must vanish on every representative of a. ℵ

Definition 1.2 (Projective $\mathcal{V}$ Operator)**.** *Let $\mathcal{S} \subseteq K[x_0, \ldots, x_n]$ be a set of polynomials. The projective vanishes set of $\mathcal{S}$ is*

$$\mathcal{V}_{\mathbb{P}}(\mathcal{S}) = \{a \in \mathbb{P}^n \mid f(a) = 0 \; \forall f \in \mathcal{S}\}$$

We say that $X \subseteq \mathbb{P}^n$ is projective variety if $X = \mathcal{V}_{\mathbb{P}}(S)$ for some set $\mathcal{S} \subseteq K[x_0, \ldots, x_n]$.

Analagously to the affine case, a projective variety is defined by a ideals, i.e.

Proposition 1.3. *If $S \subseteq K[x_0, \ldots, x_n]$ is a set of polynomials then*

$$\mathcal{V}_{\mathbb{P}}(\mathcal{S}) = \mathcal{V}_{\mathbb{P}}(\langle \mathcal{S} \rangle)$$

Proof. Let $a \in \mathcal{V}_{\mathbb{P}}(\mathcal{S})$, implies that for any polynomial $f \in \mathcal{S}$ that f vanishes on a. Let $g \in K[x_0, \ldots, x_n]$ be a polynomial. The product fg evaluated at a must also vanish. Therefore $fg \in \mathcal{S}$ hence $\mathcal{S}$ must be an ideal. The reverse inclusion comes immediately. ℵ

Further specifying, again like the affine case, we see that projective varieties are finitely generated by homogenous polynomials.

Proposition 1.4. *Any projective variety $X \subseteq \mathbb{P}^n$ is of the form $X = \mathcal{V}_{\mathbb{P}}(f_1, \ldots, f_k)$ where $f_1, \ldots, f_k \in K[x_0, \ldots, x_n]$ are homogenous polynomials.*

Proof. Let $X \subseteq \mathbb{P}^n$ be a projective variety then $X = \mathcal{V}_{\mathbb{P}}(\mathcal{S})$ where $\mathcal{S} \subseteq K[x_0, \ldots, x_n]$ is an ideal. Hilbert's Basis theorem ensures that $K[x_0, \ldots, x_n]$ is Neotherian implying there exist polynomials $f_1, \ldots, f_k \in K[x_0, \ldots, x_n]$ such that $\mathcal{S} = \langle f_1, \ldots, f_k \rangle$. Using lemma 9.16 note that if each polynomial f_i of degree d vanishes on a point $a \in \mathbb{P}^n$ then every homogenous component $f_{i,k}$ of degree k for $k \leq d$ must also vanishes on a. Since $f_i = \sum_p = 1^d f_{i,p}$ then we may further refine our list of generating polynomials. Hence any projective variety is the vanishing of a finite set of homogenous polynomials. ℵ

1.3 9.3 Projective $\mathcal{I}$-operator

It is defined essentially identically to the affine vanishing operator

Definition 1.3 (Projective $\mathcal{I}$-operator)**.** *Let $X \subseteq \mathbb{P}^n$ be a subset. The vanishing ideal of X is the set*

$$\mathcal{I}_{\mathbb{P}}(X) = \{f \in K[x_0, \ldots, x_n] \mid f(a) = 0 \; \forall a \in X\}$$

Ex: Let $X = \{[1:0]\} \subseteq \mathbb{P}^n$. Observe that for the polynomial $f(x_0, x_1) = x_1$ any representative of $[1:0]$ will always vanish. In fact we will show that $\langle x_1 \rangle = \mathcal{I}_{\mathbb{P}}(X)$. The inclusion $\langle x_1 \rangle \subseteq \mathcal{I}_{\mathbb{P}}(X)$ follows immediately as x_1 is fixed to zero. Now take $f \in \mathcal{I}_{\mathbb{P}}(X)$ then $f(a, 0) = 0$ for all $a \in K$, then we can view the polynomial as univariate in the first coordinate $f(x_0, 0)$ with infinitely many zeroes implying it must be the zero polynomial. Take a polynomial $f \in K[x_0, x_1]$ and view it as a univariate polynomial $f \in (K[x_0])[x_1]$ which has a general form

$$f = \sum_{d \geq 0} f_d(x_0) x_1^d$$

If f vanishes on X, then $f(x_0, 0) = 0$ implying that the degree zero homogenous component is also zero leaving us with an extra x_1 term

$$f = x_1 \sum_{d \geq 1} f_d(x_0) x^{d-1} \in \langle x_1 \rangle.$$

Ex: Let $X = \mathcal{V}(x_0 + x_1 - x_2) \subseteq \mathbb{P}^2$ then $\mathcal{I}(X) = \langle x_0 + x_1 - x_2 \rangle$. The ideal into vanishing ideal inclusion is immediate by definition. Now

Definition 1.4. *An ideal $I \subseteq K[x_0, \ldots, x_n]$ is a homogenous ideal if for all $f \in I$ every homogenous component of f is also in I.*

In fact we can immediately generate a more useful and straightforward characterization of a homogenous ideal.

Proposition 1.5. *An ideal $I \subseteq K[x_0, \ldots, x_n]$ is homogenous if and only if it admits a set of homogenous generators.*

Proof. Suppose that $I = \langle S \rangle$ where S is composed of all homogenous polynomials contained in I. Let $f \in I$, then $f = \sum_{k \geq 0} f_k$. Since $f_k \in \langle S \rangle$ by construction then the sum f must also be contained in $\langle S \rangle$. The other inclusion follows immediately form the fact that S is a subset of the ideal I. Now suppose that I admits a set of homogenous generators, more precisely that $I = \langle S \rangle$ where S is generated by a collection of homogenous polynomials. Let $f \in I$, then by definition

$$f = \sum_{i=1}^{m} g_i h_i$$

where $g_i \in K[x_0, \ldots, x_n]$ and $h_i \in \langle S \rangle$ of homogenous degree d_i for all i. Observe that homogenous components of f of degree k take the form

$$f_k = \sum_{i=1}^{m} (g_i h_i)_k$$

. Note that every monomial in the product $g_i h_i$ is a monomial of g_i multiplied by all of h_i. This means that every monomial of the product is of homogenous degree at least d_i. This ensures that for $k < d_i$, the homogenous component $(g_i h_i)_k = 0$. For $k \geq d_i$, monomials of homogenous degree k must be the product $(g_i)_{k-d_i} h_i$ as every monomial of the product $g_i h_i$ is a product of a monomial of g_i with h_i. This implies that $(g_i h_i)_k \in I$ then f_k is a sum of elements of I implying that it must be contained in I. Therefore I must be a homogenous ideal. ℵ

With this we can give a characterization of not just ideals, but vanishing ideals.

Proposition 1.6. *If $X \subseteq \mathbb{P}^n$ is any subset, then $\mathcal{I}(X)$ is a homogenous radical ideal.*

Proof. We begin by proving that $\mathcal{I}(X)$ is a radical ideal. Suppose there exists an integer m such that $f^m \in \mathcal{I}(X)$, then for a point $a \in X$

$$0 = f^m(a) = (f(a))^m.$$

Since K is an infinite field it has no zero divisors implying that $f(a) = 0$ and that $f \in \mathcal{I}(X)$. Hence $\mathcal{I}(X)$ is a radical ideal. Since f vanishes on a then every homogenous component of f must also vanish on a and is therefore contained in $\mathcal{I}(X)$ implying that it is homogenous. ℵ

With this we go on to establish identical properties to those of affine varieties related to the existence of unique irreducible decompositions of projective varieties. The proofs themselves follow exactly as in the affine case.

1.4 9.4 Affine Restrictions

We saw in the very first section of this chapter that a projective space $\mathbb{P}^n$ could be viewed as $\mathbb{A}^n \sqcup \mathbb{P}^{n-1}$. Quite reasonably we ask whether a projective variety could be stratified in this way. Consider the projective variety $X = \mathcal{V}_{\mathbb{P}}(x_0^2 + x_1^2 - x_2^2) \subseteq \mathbb{P}^2_{\mathbb{C}}$. we saw that the affine component of a projective space is the collection of points where the first coordinate is non-zero or more succinctly when the first coordinate was fixed at 1. Similarly, fix $x_0 = 1$, then we have a subvariety

$$X_0 = \mathcal{V}_{\mathbb{A}}(1 + x_1^2 - x_2^2) \subseteq X$$

whose solutions are entirely contained in $\mathbb{A}^2_{\mathbb{C}}$. We give this phenomenon a more precise definition:

Definition 1.5 (Affine Patch)**.** *For each $i \in \{0, \dots, n\}$ the i-th affine patch of $\mathbb{P}^n$ is the set*

$$\mathbb{A}_i^n = \{[a_0 : \cdots : a_n] \in \mathbb{P}^n \mid a_i \neq 0\}$$

and for any set $X \subseteq \mathbb{P}^n$, the intersection $X \cap \mathbb{A}_i^n$ is called the i-th affine restriction of X.

Of course by simply dropping the i-th component, which is fixed at 1, we are in bijection with $\mathbb{A}^n$. Then the i-th restriction of a projective variety $X = \mathcal{V}_{\mathbb{P}}(\mathcal{S})$ is $\mathcal{V}_{\mathbb{A}}(\mathcal{S}_i)$ where

$$\mathcal{S}_i = \{f(x_0, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_n) \mid f \in \mathcal{S}\}$$

1.5 9.5 Projective Closure

In the previous section we developed a way to take the global projective variety and investigate restriction to local affine patches. Now we aim to develop a method to take some affine patches and join them together into a projective variety. Notice though that in the process of restricting to affine patche lose the points at infinity. So if we were to start with affine patches and attempt to move back into a projective variety, we would need to devise a way to add point at infinity and ensure that the resulting set is actually a projective variety. To do this we define the projective closure.

Definition 1.6 (Projective Closure)**.** *Let $X \subseteq \mathbb{A}^n$ be an affine variety, and let $j_0(X)$ be the image of X in the first affine patch $\mathbb{A}_0^n \subseteq \mathbb{P}^n$. The projective closure of X, denoted $\overline{X}$ is the intersection fo all projective varieties that contain $j_0(X)$.*

Remark 1.3. Observe that, under the zariski topology, this is exactly the definition of a closed set.

- Consider the affine variety $\mathcal{V}_{\mathbb{A}}(1 + x_1 - x_2) \subseteq \mathbb{A}^2$ which has the vanishing set $\{(a, 1 + a) : a \in K\}$. Under the josure, j_0 we can embed this variety as a set in $\mathbb{P}^2$ defined as

  $$j_0(X) = \{[1 : a : a + 1] : a \in K\} \subseteq \mathbb{P}^2.$$

  Observe that $[1 : a : a+1] = [\frac{1}{a} : 1 : 1 + \frac{1}{a}]$ means that a must be furthe resitrcted to $a \in K \setminus \{0\}$. Take $b = \frac{1}{a}$ then we can rewrite the elements of $j_0(X)$ to be $[b : 1 : 1 + b]$. Then in terms of our polynomial, for every point in $j_0(X)$, $f(b, 1, 1 + b) = 0$ can be viewed as a univariate polynomial $\tilde{f}(x, 1, 1 + x) = 0$ which means that $\tilde{f}$ is the univariate polynomial. If we were to treate $j_0(X)$ as a projective variety then we have a univariate polynomial vanishing on $\mathbb{P}^2 = \mathbb{A}^2 \sqcup \mathbb{P}^1$. In particular this polynomial is vanishing on $\mathbb{P}^1$ which in this case is only the poin $[0 : 1]$ then $f(0, 1, 1)$ must be a solution so we have the point $[0 : 1 : 1]$ as a solution. So we determine that we must augment $j_0(X)$ as $j_0(X) \cup \{[0 : 1 : 1]\}$. Note that we STILL don't yet know whether this is a projective variety because restricting to the first affine path agrees with $j_0(X)$ but we don't know how this variety is changed when restricting to different affine patches. This means that we must determine the polynomial that vanishes and luck for us there is one that I am too lazy to compute so I will take from the book:

  $$\overline{X} = \mathcal{V}_{\mathbb{P}}(x_0 + x_1 - x_2)$$

- Let $\mathcal{V}_{\mathbb{A}}(x_2 + x_1^2) \subseteq \mathbb{A}^2$ has josure

  $$j_0(X) = \{[1 : a : a^2] | a \in K\}$$

  which under $b = \frac{1}{a}$ can be viewed as the set of points $\{[b^2 : b : 1] : b \in K \setminus \{0\}\}$. Similarly we must add a single point and then determine whether this point with the josure is actually the vanishing of the polynomial. It is hence $\overline{X} = \mathcal{V}_{\mathbb{P}}(x_0 x_1 - x_2^2)$.

Observe that the resulting polynomial of the variety is now of some homogenous degree. In fact, as we will state soon, the projective closure of an affine variety is always the "homogenization" of the ideal generating the affine variety. This gives us a much simpler characterization of the projective closure than the rather abstract intersection of all varieties containing the josure.

Definition 1.7. *Let $f \in K[x_1, \dots, x_n]$ be a polynomial of degree d. The homogeniztion of the polynomial is define as*

$$\overline{f} = x_0^d f(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}) \in K[x_0, \dots, x_n].$$

Observe that if we were to look at the josure of a homogenized polynomial, that is we evaluate $\overline{f}(1, x_1, \ldots, x_n)$ we will get back our unhomogenized polynomial f. Reframing our previous examples we would like to determine whether if $X = \mathcal{V}_{\mathbb{A}}(f_1, \ldots, f_k)$ that $\overline{X} = \mathcal{V}_{\mathbb{P}}(\overline{f}_1, \ldots, \overline{f}_k)$ yet the accursed twisted cubic will laugh in our face. I don't feel like walking through this example so I will make some general remarks about it insead. The affine twisted cubic $\mathcal{V}_{\mathbb{A}}(x_2 - x_1^2, x_3 - x_1^3)$ will have a josure $j_0(X) = \{[1 : a : a^2 : a^3] \mid a \in K\} \subseteq \mathbb{P}^3$ and to make it a projective variety we only need to add the point $[0 : 0 : 0 : 1]$. If we were to look at a variety definined by the homogenization of the polynomials then we would arrive at $j_0(X) \sqcup \{0 : 0 : b : c\}$, i.e. we would add too many points. Instead we actually another defining polynomial rather than simply homogenizing the defining polynomials of the affine variety. So instead of homogenizing some polynomials we will actually just homogenize the entire ideal.

Definition 1.8 (Homogenization of a Set)**.** *Let $\mathcal{S} \subset K[x_1, \ldots, x_n]$ be a set of polynomials. The homogenization of S is the set*

$$\overline{\mathcal{S}} = \{\overline{f} : f \in \mathcal{S}\} \subseteq K[x_0, \ldots, x_n]$$

One thing to note is that generally the homogenization of an ideal is not also an ideal. If we were to take a product $g\overline{f}$ where $g \in K[x_0, \ldots, x_n]$ and inhomogenous and $\overline{f} \in \overline{\mathcal{S}}$ then our product is not homogenous and so can not be in $\overline{\mathcal{S}}$. With the homogenization of a set we get the right information to recharacterize the affine closure in a way that plays better with the vanishing and ideal operators.

Proposition 1.7. *If $X \subseteq \mathbb{A}^n$ and affine variety then $\overline{X} = \mathcal{V}_{\mathbb{P}}(\overline{\mathcal{I}_{\mathbb{A}}(X)}) \subseteq \mathbb{P}^n$.*

Proof. ($\subseteq$) Let $[1 : a_1 : \cdots : a_n] \in j_0(X)$ and $g \in \overline{\mathcal{I}_{\mathbb{A}}(X)}$ then $g = \overline{f}$ for some $f \in \mathcal{I}_{\mathbb{A}}(X)$ and $g(1, a_1, \ldots, a_n) = f(a_1, \ldots, a_n) = 0$ then $j_0(X)$ is contained in $\mathcal{V}_{\mathbb{P}}(\overline{\mathcal{I}_{\mathbb{A}}(X)})$. Since $\overline{X}$ is the smallest projective variety to contain $j_0(X)$ then $\overline{X}$ must be contained in $\mathcal{V}_{\mathbb{P}}(\overline{\mathcal{I}_{\mathbb{A}}(X)})$. ($\supseteq$) It suffices to show that

$$\mathcal{I}_{\mathbb{P}}(\overline{X}) \subseteq \mathcal{I}_{\mathbb{P}}(\mathcal{V}_{\mathbb{P}}(\overline{\mathcal{I}_{\mathbb{A}}(X)}))$$

because application of the projective vanishing operator will give us the desired containment. Suppose that $g \in \mathcal{I}_{\mathbb{P}}(\overline{X})$, as g is contained in a homogenous ideal it must also be homogenous. The homogenous polynomial g is not necessarily the homogenization of a polynomial so to force this condition let $k \in \mathbb{N}$ be the maximal choice such that $g = x_0^k h$ for some $h \in K[x_0, \ldots, x_n]$. Observe that forcing k to be maximal implies that from at least one monomial of g we must be able to factor the entirety of an x_0 component out, if this weren't the case then there would be some x_0 terms in every monomial and k would not be maximal. Since g is homogenous of some degree d, then this monomial has degree $d - k$. Note that this monomial must have minimal degree after factoring as any term with some x_0 has less remaining degree distributed over terms x_i where $i \neq 0$. Take $h_0 = h(1, x_1, \ldots, x_n)$ and observe that this removes the x_0 term from every monomial and utilizing the conlusion of the previous two sentences this means that the monomial of degree $d - k$ is now of maximal degree in h_0. The $\overline{h_0} = x_0^{d-k} h_0$ which returns us back to the polynomial h. Since g vanishes on all of $\overline{X}$ then it must vanish on all of $j_0(X)$, in particular this means that g must vanish on all of X. Since

$$h_0(a_1, \ldots, a_n) = h(1, a_1, \ldots, a_n) = g(1, a_1, \ldots, a_n) = 0$$

we see that $h_0 \in \mathcal{I}_{\mathbb{A}}(X)$ and by definition $h \in \overline{\mathcal{I}_{\mathbb{A}}(X)}$ and that $g \in \langle \overline{\mathcal{I}_{\mathbb{A}}(X)} \rangle$. Then

$$g \in \langle \overline{\mathcal{I}_{\mathbb{A}}(X)} \rangle \subseteq \mathcal{I}_{\mathbb{P}}(\mathcal{V}_{\mathbb{P}}(\overline{\mathcal{I}_{\mathbb{A}}(X)}))$$

and we are done. ℵ

While we can't in general conclude that $\overline{\mathcal{V}_{\mathbb{A}}(f_1, \ldots, f_k)} = \mathcal{V}_{\mathbb{P}}(\overline{f_1}, \ldots, \overline{f_k})$ we get a special case when $i = 1$.

Proposition 1.8. *If $f \in K[x_0, \ldots, x_n]$ then $\overline{\mathcal{V}_{\mathbb{A}}(f)} = \mathcal{V}_{\mathbb{P}}(\overline{f})$.*

Proof. Let $X = \mathcal{V}_{\mathbb{A}}(f)$, the Nullstelensatz says that $\mathcal{I}_{\mathbb{A}}(\mathcal{V}_{\mathbb{A}}(X)) = \langle q_1 \cdots q_\ell \rangle$ where $f = q_1^{\alpha_1} \cdots q_\ell^{\alpha_\ell}$ is an irreducible decomposition of f. Using the previous proposition $\overline{\mathcal{V}_{\mathbb{A}}(X)} = \mathcal{V}_{\mathbb{P}}(\overline{\mathcal{I}_{\mathbb{A}}(X)}) = \mathcal{V}_{\mathbb{P}}(\overline{\langle q_1 \cdots q_\ell \rangle})$. It remains to show that this is equivalent to $\mathcal{V}_{\mathbb{P}}(\overline{f})$. Let $f \in \langle q_1 \cdots q_\ell \rangle$ then $\overline{f} \in \overline{\langle q_1 \cdots q_\ell \rangle}$ which implies that

$$\mathcal{V}_{\mathbb{P}}(\overline{f}) \supseteq \mathcal{V}_{\mathbb{P}}(\overline{\langle q_1 \cdots q_\ell \rangle}).$$

Now suppose that $a \in \mathcal{V}_{\mathbb{P}}(\overline{\langle q_1 \cdots q_\ell \rangle})$ then for some $\overline{f} \in \overline{\langle q_1 \cdots q_\ell \rangle}$, $\overline{f}(a) = 0$. Note that $\overline{f} = \overline{q_1}^{\alpha_1} \cdots \overline{q_\ell}^{\alpha_\ell}$. The fact that $\overline{f}$ vanishes on a implies that $\overline{q_i}^{\alpha_i}$ vanishes on a for some i. Then for any $g \in \overline{\langle q_1 \cdots q_\ell \rangle}$ it follows that g must also vanish on a because it will have the vanishing homogenized q_i component. ℵ

One thing that we may have noticed in the examples working through projective closures was the fact that we never seemed to be adding affine points, only points at infinity. In fact these are examples of the following proposition.

Proposition 1.9. *The affine restriction of the projective closure of an affine variety X is always X. $(\overline{X})_0 = X$.*

Proof. Let $X \subseteq \mathbb{A}^n$ be an affine variety. Define $\mathcal{S} = \overline{\mathcal{I}_{\mathbb{A}}(X)}$ and note that $\overline{X} = \mathcal{V}_{\mathbb{P}}(\mathcal{S})$. Then the affine restriction $(\overline{X})_0$ vanishes on a set of polynomials

$$\mathcal{S}_0 = \{g(1, a_1, \ldots, a_n) \mid g \in \mathcal{S}\} = \{f(a_1, \ldots, a_n) \mid f \in \mathcal{I}_{\mathbb{A}}(X)\}$$

where the second equality comes from the fact that g must be the homogenization of some polynomial f and evaluating g at $x_0 = 1$ simply returns us to evaluation f at $a_1, \ldots, a_n$. The final set is just $\mathcal{I}_{\mathbb{A}}(X)$ then $\mathcal{V}_{\mathbb{A}}(\mathcal{I}_{\mathbb{A}}(X)) = X$ and we are done. ℵ

1.6 9.6 Project Nullstelensatz

If we are to take inspiration from the affine NS then we aim to prove that

$$\mathcal{I}_{\mathbb{P}}(\mathcal{V}_{\mathbb{P}}(I)) = \sqrt{I}$$

for some ideal $I \in K[x_0, \ldots, x_n]$. But we have a problem here if $I = \langle x_0, x_1, \ldots, x_n \rangle$. The only point that vanishes on a polynomial in here would be $[a_0 : \cdots : a_n]$ where each $a_i = 0$. Since there is no such point in $\mathbb{P}^n$ then $\mathcal{V}_{\mathbb{P}}(I) = \emptyset$. Yet every polynomial vanishes on the emptyset hence $\mathcal{I}_{\mathbb{P}}(\mathcal{V}_{\mathbb{P}}(I)) = K[x_0, \ldots, x_n] \neq I$ where we note that I is already radical. So with this we aim to disregard this annoying ideal.

Definition 1.9. *An ideal $I \subseteq K[x_0, \ldots, x_n]$ is irrelivant if $I \supseteq \langle x_0, \ldots, x_n \rangle$. Otherwise I is called relevant.*

With this the projective NS is the affine NS reframed with these classification.

Proposition 1.10. *Let $I \subseteq K[x_0, \ldots, x_n]$ be a homogenous ideal.*

1. *$\mathcal{V}_{\mathbb{P}}(I) = \emptyset$ if and only if $\sqrt{I} \supseteq \langle x_0, \ldots, x_n \rangle$.*
2. *If $\sqrt{I}$ is relevant then $\mathcal{I}_{\mathbb{P}}(\mathcal{V}_{\mathbb{P}}(I)) = \sqrt{I}$.*

But before proving this we will utilize a different way to move between affine and projective varieties through cones.

Definition 1.10. *Let $X \subseteq \mathbb{P}^n$ be a subset. The affine cone over X is the set*

$$C(X) = \{(0, \ldots, 0)\} \cup \{(a_0, \ldots, a_n) \mid [a_0 : \cdots : a_n] \in X\} \subseteq \mathbb{A}^{n+1}$$

With the affine cone in mind we'll prove the following proposition that allow interchagnes between the affine and projecive vanishng and ideal operators.

Proposition 1.11. *Let $I \subsetneq K[x_0, \ldots, x_n]$ be a homogenous ideal then*

$$\mathcal{V}_{\mathbb{A}}(I) = C(\mathcal{V}_{\mathbb{P}}(I))$$

Proof. Since we are looking to show that $\mathcal{V}_{\mathbb{A}}(I)$ is a cone then we must first show that for any homogenous ideal that this variety admits $(0, \ldots, 0)$. Suppose that it did not admit the origin, this means that there exsists a homogenous polynomial $f \in I$ such that $f(0, \ldots, 0) \neq 0$. This is only possible if the polynomial is non-zero and constant which forces I to admit such a generator. But if it were to admit such a generator then $I = K[x_0, \ldots, x_n]$ contradicting our choice of I. Since each polynomial in I is homogenous, if it were to vanish on a point $(a_0, \ldots, a_n)$ then it must vanish on any representative $[a_0 : \cdots : a_n]$ and vice versa hence

$$(a_0, \ldots, a_n) \in \mathcal{V}_{\mathbb{A}}(I) \iff [a_0 : \cdots : a_n] \in \mathcal{V}_{\mathbb{P}}(I) \iff (a_0, \ldots, a_n) \in C(\mathcal{V}_{\mathbb{P}}(I))$$

therefore we are done. ℵ

Proposition 1.12. *Let $X \subseteq \mathbb{P}^n$ be a non-empty subset. Then*

$$\mathcal{I}_{\mathbb{P}}(X) = \mathcal{I}_{\mathbb{A}}(C(X)).$$

Proof. Note that $\mathcal{I}_{\mathbb{P}}(X)$ is a non-empty homogenous ideal which means that none of the generating polynomials can be constant. Then on any point $(0,\dots,0) \in \mathbb{A}^{n+1}$ a polynomial in this projective vanishing ideal might vanish. Similarly the fact that a polynomial vanishes on the projective variety X means that it must vanish on it's affine restriction. Then we can redifine the projective vanishing ideal as

$$\mathcal{I}_{\mathbb{P}}(X) = \{f \in K[x_0,\dots,x_n] \mid f(a_0,\dots,a_n) = 0 \text{ if } [a_0 : \cdots : a_n] \in X, f(a_0,\dots,a_n) = 0 \text{ if } (a_0,\dots,a_n) = (0,\dots,0)\} \tag{3}$$

$$= \mathcal{I}_{\mathbb{A}}(\{0,\dots,0\} \cup \{(a_0,\dots,a_n) \mid [a_0 : \cdots : a_n] \in X\}) \tag{4}$$

$$= \mathcal{I}_{\mathbb{A}}(C(X)) \tag{5}$$

ℵ

Now that we have the ability to move between projective and affine varieties and vanishing ideals through cones we can prove the projective NS.

Proof. (a) Suppose that $\sqrt{I} \supseteq \langle x_0,\dots,x_n\rangle$. Note that if $a \in \mathcal{V}_{\mathbb{P}}(I)$ then there exits some polynomial $f \in I$ such that $f(a) = 0$ then $f \in \sqrt{I}$ hence $a \in \mathcal{V}_{\mathbb{P}}(\sqrt{I})$. Suppose that $a \in \mathcal{V}_{\mathbb{P}}(\sqrt{I})$ then for some $m > 0$ there exists an $f^m \in \sqrt{I}$ such that $f^m(a) = 0$. By definition it follows that $f \in I$ and as $f^m(a) = f^{m-1}(a)f(a) = 0$ it follows that $f(a) = 0$. Therefore $\mathcal{V}_{\mathbb{P}}(I) = \mathcal{V}_{\mathbb{P}}(\sqrt{I})$. Then

$$\mathcal{V}_{\mathbb{P}}(I) = \mathcal{V}_{\mathbb{P}}(\sqrt{I}) \subseteq \mathcal{V}_{\mathbb{P}}(\langle x_0,\dots,x_n\rangle) = \emptyset.$$

Now suppose that $\mathcal{V}(I) = \emptyset$, then either $I = K[x_0,\dots,x_n]$ in which case $\sqrt{I} = I \supseteq \langle x_0,\dots,x_n\rangle$. Suppose that $I \subsetneq K[x_0,\dots,x_n]$. Utilizing prop 1.11

$$\mathcal{V}_{\mathbb{A}}(I) = C(\mathcal{V}_{\mathbb{P}}(I)) = C(\emptyset) = \{(0,\dots,0)\}.$$

Under the affine NS

$$\sqrt{I} = \mathcal{I}_{\mathbb{A}}(\mathcal{V}_{\mathbb{A}}(I)) = \mathcal{I}_{\mathbb{A}}(\{0,\dots,0\}) = \langle x_0,\dots,x_n\rangle$$

(b) Now assume that $\sqrt{I}$ is relevant. Suppose that $\mathcal{V}_{\mathbb{P}}(I) = \emptyset$. The first case tells us that $\sqrt{I} \supseteq \langle x_0,\dots,x_n\rangle$ but the only relevant ideal that can contain the maximal irrelevant ideal is the full ring i.e. $\sqrt{I} = K[x_0,\dots,x_n]$. Then $\mathcal{I}_{\mathbb{P}}(\mathcal{V}_{\mathbb{P}}(I)) = \mathcal{I}_{\mathbb{P}}(\emptyset) = K[x_0,\dots,x_n]$ and we are done. Now suppose that $\mathcal{V}_{\mathbb{P}}(I) \neq \emptyset$ which means that $I \subsetneq K[x_0,\dots,x_n]$. Then,

$$\mathcal{I}_{\mathbb{P}}(\mathcal{V}_{\mathbb{P}}(I)) = \mathcal{I}_{\mathbb{A}}(C(\mathcal{V}_{\mathbb{P}}(I))) = \mathcal{I}_{\mathbb{A}}(\mathcal{V}_A(I)) = \sqrt{I}$$

hence we are done.

ℵ

Remark 1.4. $\mathcal{V}_{\mathbb{P}}(f_1,\dots,f_k) = \emptyset$ if and only if there exists d such that every monomial of degree d is contained in $\langle f_1,\dots,f_k\rangle$.

Proof. SKETCH : ($\Longleftarrow$) If every monomial of degree d is contained in $\langle f_1,\dots,f_k\rangle$ then in particular we contain $x_0^d,\dots,x_n^d$ on which means that we contain the irrelevant ideal and projective NS tells us this vanishing is empty. ($\Longrightarrow$) If the vanishing is empty then the projective NS tells use that $\sqrt{\langle f_1,\dots,f_k\rangle} \supseteq \langle x_1,\dots,x_n\rangle$. By definition of radical ideals there exist a d_i such that $x_i^{d_i}$ is contained in the radical of the vanishing ideal. Now take $N = \max\{d_1,\dots,d_n\}$ which ensures that any x_i^N is contained in the vanishing ideal. Let $d = (n+1)N$ ℵ

2 CH 10 Maps of Projective Varieties

2.1 10.1 Regular Maps of Projective Varieties

In the affine case we established that polynomials maps between affine varieties maps that are polynomial in each coordinate from one variety into another. In projective varieties this perspective makes less sense because of the scaling factor that acts as an equivalence among projective points. So instead we consider homogenous maps between projective varieties, we have to be careful though because we may accidentally vanish on a point in the variety and end up with a function that isn't well defined. To avoid this we define maps between projective varieties as follows:

Definition 2.1. *Let $X \subseteq \mathbb{P}^m$ and $Y \subseteq \mathbb{P}^n$ be projective varieites. A map $F : X \to Y$ is called polynomial if there exists polynomials $f_0, \dots, f_n \in K[x_0, \dots, x_m]$ homogenous of the same degree, such that $X \cap \mathcal{V}(f_0, \dots, f_m) = \emptyset$ and*

$$F(a) = [f_0(a) : \cdots : f_n(a)] \; \forall a \in X$$

If we can take f_i to be linear in all i then we call the map between varieties linear.

Unfortunately the requirement that $X \cap \mathcal{V}(f_0, \dots, f_m)\emptyset$ tends to be far too restrictive. The restrictiveness arises when we try and define polynomial maps over all varieities, this intersection requirement cuts out many obvious maps and as we will see later polynomials maps on projective varieties are always constant. Lets illustrate this with an example, take $X = \mathcal{V}(y^2 - xz) \subseteq \mathbb{P}^2$ and consider the function $x, y \in K[x, y, z]$. We can use these function to define a polynomial map as $X \cap \mathcal{V}(x, y) = \{[0 : 0 : 1]\}$ but we still get a map from a subset of X to $\mathbb{P}^1$ which simply excludes this point:

$$[x : y] : X \setminus \mathcal{V}(x, y) \to \mathbb{P}^1 \tag{6}$$

$$[a : b : c] \mapsto [a : b] \tag{7}$$

Now if we consider polynomials $y, z \in K[x, y, z]$ notice that we again don't get a polynomial map but we get a regular map now that excludes $[1 : 0 : 0]$. Notice that these maps together cover the entirety of X so we can define a map

$$F : X \to \mathbb{P}^1 \tag{8}$$

$$[a : b : c] \mapsto \begin{cases} [a : b], & [a : b : c] \notin \mathcal{V}(x, y) \\ [b : c], & [a : b : c] \notin \mathcal{V}(y, z) \end{cases} \tag{9}$$

In fact for any projective variety X and every point $p \in X$ there exist a pair of possibly different polynomials f, g such that $p \notin \mathcal{V}(f, g)$ meaning we can define a map over the entire projective variety by constructing a piecewise map for every p. This motivates the definition for a regular map:

Definition 2.2. *A map $F : X \to Y$ of projective varieties is called regular if for every $p \in X$, there exists polynomials $f_0, \dots, f_n \in K[x_0, \dots, x_m]$ homogenous of the same degree such that $p \notin \mathcal{V}(f_0, \dots, f_n)$ and*

$$F(a) = [f_0(a) : \cdots : f_n(a)] \; \forall a \in X \setminus \mathcal{V}(f_0, \dots, f_n)$$

It's important to note that every polynomial map is a regular map, in the sense that we were able to find a single polynomial for all points in the projective variety, but not every regular map is polynomial.

Remark 2.1. Any regular map $F : \mathbb{P}^1 \to \mathbb{P}^n$ is a polynomial map.

Proof. Let F be a regular map from $\mathbb{P}^1$ to $\mathbb{P}^n$. This implies that for any point $p \in \mathbb{P}^1$ there exists polynomial $f_0, \dots, f_n \in K[x, y]$ homogenous of the same degree such that $p \notin \mathcal{V}(f_0, \dots, f_n)$ and that

$$F(a) = [f_0(a) : \cdots : f_n(a)]$$

for all $a \in \mathbb{P}^1 \setminus \mathcal{V}(f_0, \dots, f_n)$. If $\mathbb{P}^1 \cap \mathcal{V}(f_0, \dots, f_n) = \emptyset$ then we are done. Assume that the intersection is non-empty then there exists a point $p \in \mathbb{P}^1$ such that $f_i(p) = 0$ for all i. If $p = [0 : a]$ for some $a \in K \setminus \{0\}$ then we may view each f_i as a univariate polynomial vanishing at a. Since our field is algebraically closed then $f_i(x, y) = (y - a)g_i(x, y)$ allowing us to choose a new collection $g_0, \dots, g_n$

which doesn't vanish at p, is equivalent upon evaluation as point in $\mathbb{P}^n$, and that only add p to the domain of F. If $p = [1 : a]$ for some $a \in K$, then we can do the same process to product a collection on which this point doesn't vanish. Once we have a collection that no longer vanishes at p, we repeat the process by checking for projective points on which this new collection doesn't vanish. Note that any polynomial only has a finite number of solutions ensureing that this process terminates at some collection $h_0, \ldots, h_n$. In particular this process must terminate a point where no polynomial has a common root or we have forced a 1 in some component in both cases implying the vanishing set is empty. ℵ

2.2 10.2 Projective Equivalences

Of course now that we have developed the notion of a map between projective varieties. This naturally asks us to consider when two projective varieties are isomorphic to one another. In particular, if for projective varieties $X \subseteq \mathbb{P}^m$ and $Y \subseteq \mathbb{P}^n$ there exist a pair of regular maps $F : X \rightleftarrows Y : G$ where $F \circ G = \mathrm{id}_X$ then we say these varieites are isomorphic. Verifying an isomorphism has no special conditions other than ensuring that given a map F, that the inverse that you define is a well defined regular map on Y. We are particularly interested in a very special isomorphism of $\mathbb{P}^n$ to $\mathbb{P}^n$. Similar to affine space we can identify $\mathbb{P}^n$ with a vector space, note that we could never really be a vector space because we never have the origin, and consider for a point $[a_0 : a_1 : a_2] \in \mathbb{P}^2$

$$\begin{bmatrix} a_0 & a_1 & a_2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix}$$

giving us a linear map

$$[a_0 : a_1 : a_2] \mapsto [a_0 + a_1 : a_0 + 2a_2 : a_1 - a_2].$$

More generally for a point $[a_0 : \cdots : a_n]$ and a $(n+1) \times (n+1)$ invertible matrix we get a linear map

$$F_A = \left[\sum_{j=0}^{n} A_{0,j} x_j : \cdots : \sum_{j=0}^{n} A_{n,j} x_j \right]$$

that, given the invertibility of A, has an inverse $F_{A^{-1}}$. This then implies that F_A and $F_{A^{-1}}$ are enough information to provide an isomorphism of projective n space to itself.

Definition 2.3 (Projective Equivalence)**.** *A projective equivalence of $\mathbb{P}^n$ is an isomorphism of the form*

$$F_A : \mathbb{P}^n \to \mathbb{P}^n \tag{10}$$

$$p \mapsto Ap \tag{11}$$

where A is an $(n+1) \times (n+1)$ matrix with entries in K. We say that projective varieties $X, Y \subseteq \mathbb{P}^n$ are projectively equivalent if there exists a projectively equivalent if there exists a projective equivalence such that $F_A(X) = Y$.

Remark 2.2. If X is a projective variety then a projective equivalence always maps $X = \mathcal{V}(f_1, \ldots, f_k)$ to a projective variety $F_A(X)$. To see this consider that

$$F_A(X) = \{F(a) \in \mathbb{P}^n : f_i(a) = 0 \ \forall i\} = \{a \in \mathbb{P}^n : (f_i \circ F_A^{-1})(a) = 0 \ \forall i\}$$

where $f_i \circ F_A^{-1}$ is polynomial because it is just a composition of two polynomial. Then it's immediate to verify that $(f_i \circ F_A^{-1})(F_A(a)) = f_i(a) = 0$. This assertion is generally true but requires us to develop some more machinery and will be tackled in Theorem 11.65.

The language of projective equivalences, on a degree two polynomial, exhibits a remarkable level of regularity. That is

Theorem 2.3. *Let $f, g \in K[x, y, z]$ be irreducible and homogenous of degree 2. Then $\mathcal{V}(f)$ and $\mathcal{V}(g)$ are projectively equivalent in $\mathbb{P}^2$.*

Proof. Choose two distinct points $[a_1 : b_1 : c_1], [a_2 : b_2 : c_2] \in \mathcal{V}(f)$, and the matrix

$$\begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \\ c_1 & c_2 \end{bmatrix}$$

must be of rank 2. If every point of $\mathcal{V}(f)$ satisfies the linear equations given by this matrix then we have $\mathcal{V}(f)$ to be contained in the vanishing set of a linear function. But the radical of the ideal generated by a linear polynomial is just the same ideal which implies that we were able to express a quadratic of homogenous degree 2 as the product of a linear polynomial. This contradiction implies that there must exist a third point $[a_3 : b_3 : c_3]$ such that

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

is off full rank and therefore invertible. Therefore A induces a projective equivalence F_A defined by A maps $[1 : 0 : 0], [0 : 1 : 0], [0 : 0 : 1]$ to our three points chosen from $\mathcal{V}(f)$. The inverse identifies $\mathcal{V}(f)$ with some irreducible conic $\mathcal{V}(g)$ such that those three points are contained in said irreducible conic. But a homogenous polynomial of degree 2 that vanishes on those three points has the form

$$g = \alpha xy + \beta yz + \gamma xz.$$

All α, β, and γ must be non-zero or g would be reducible contradiction our assertion that $\mathcal{V}(g)$ is irreducible. Then the projective equivalence must be a map

$$[a : b : c] \mapsto [\alpha^{-1}a : \beta^{-1}b : \gamma^{-1}c]$$

which identifies $\mathcal{V}(g)$ with projective variety $\mathcal{V}(xy + xz + yz)$. Then composing these we see that we have a projective equivalence. ℵ

From this we can actually make a rather powerful isomorphism statememt:

Theorem 2.4. *Let $f \in K[x, y, z]$ be an irreducible, homogenous polynomial of degree 2. Then $\mathcal{V}(f) \cong \mathbb{P}^1$.*

Becuase we know that all varieties of this form are isomorphic to one another it just suffices to show that one of these is isomorphic to $\mathbb{P}^1$. Refering to $\mathcal{V}(y^2 - xz)$ we can show this and the result of the proof follows.

2.3 10.3 Veronese Maps

Veronese maps are special because they give us a mechanism to identify, under projective equivalence, any variety defined by complex polynomials of high homogenous degree with varieties defined by a collection of polynomials of degree at most two. Using this result we can then establish that any polynomials map of projective varieties is linear(we saw something similar to this argument in 10.1.6).

Definition 2.4 (Veronese Maps)**.** *For any $d, n \geq 1$, the Veronese map $F_{d,n}\mathbb{P}^n \to \mathbb{P}^{\binom{d+n}{k}-1}$ is the polynomial map*

$$F_{d,n} = [x_0^d : x_0^{d-1}x_1 : x_0^{d-1}x_2 : \cdots : x_n^d]$$

where the monomials appearing in the definition are all of the possible monomials of degree d in the variables $x_0, \ldots, x_n$.

Note that we can view $F_{d,n}$ as a regular map because of the manner in which we defined it. That is let $\Delta = \mathcal{V}(x_0^d, x_0^{d-1}x_1, x_0^{d-1}x_2, \ldots, x_n^d)$ and observe that if a point $p \in \mathbb{P}^n \cap \Delta$ then the componentes $p_i^d = 0$ for all i which means that all $p_i = 0$ but a point with zeroes in each component can not be contained in $\mathbb{P}^n$. Since $F_{d,n}$ is polynomial it is by definition regular. Let $D = (d_0, \ldots, d_n)$ be the tuples which index the degree $d_0 + \cdots + d_n = d$ monomials. For example

$$X^D = x_0^{d_0} \cdots x_n^{d_n}$$

and we will call the cordinates of $\mathbb{P}^{\binom{d+n}{k}-1}$ by y_D so that $F_{d,n}$ is the polynomial map whose y_D-coordinate is x^D.

Proposition 2.1. *For any $d, n \geq 1$, the image of the Veronese map $F_{d,n} : \mathbb{P}^n \to \mathbb{P}^{\binom{d+n}{k}-1}$ is*

$$X_{d,n} = \mathcal{V}(\{y_D y_{D'} - y_E y_{E'} \mid D + D' = E + E'\}) \subseteq \mathbb{P}^{\binom{d+n}{k}-1}$$

Proof. Let $a \in F_{n,d}(\mathbb{P}^n)$ then for any D, D', E, E' such that $D + D' = E + E'$ given by the vanishing

$$a^D a^{D'} - a^E a^{E'} = (a_0^{d_0} \cdots a_n^{d_n})(a_0^{d'_0} \cdots a_n^{d'_n}) - (a_0^{e_0} \cdots a_n^{e_n})(a_0^{e'_0} \cdots a_n^{e'_n}) = 0$$

then we see immediately that $F_{d,n}(\mathbb{P}^n) \subseteq X_{d,n}$. Now take $a \in X_{d,n}$ then it follows by the conditions on the vanishing that for tuples D, D', E, E' that $a_D a_{D'} = a_E a_{E'}$. This can be immediately extended to show that for any D_k, E_k where $D_1 + \cdots + D_k = E_1 + \cdots + E_k$ that

$$a_{D_1} \cdots a_{D_k} = a_{E_1} \cdots a_{E_k}.$$

In particular for $D = (d_0, \ldots, d_n)$ we have that

$$a_D^d = (a_0^{d_0} \cdots a_n^{d_n})^d = a_{(d,0,\ldots,0)}^{d_0} \cdots a_{(0,\ldots,d)}^{d_n}.$$

Note that at least one coordinate must be non-zero because each index a_i comes from a component of a projective class in $\mathbb{P}^n$. WLOG suppose that $a_{(d,\ldots,0)} \neq 0$ then we can rescale this projective component by this component so that $a_{(d,\ldots,0)} = 1$. For $1 \leq i \leq n$ define

$$b_i = b_{((d-1),\ldots,1,\ldots,0)}$$

where the 1 lives in the i-th index of the tuple. Then

$$\begin{aligned}
F_{d,n}(b) &= b_0^{d_1} \cdots b_n^{d_n} && (12)\\
&= a_{(d,0,\ldots,0)}^{d_0} a_{(d-1,1,\ldots,0)}^{d_1} \cdots a_{(d-1,0,\ldots,1)}^{d_n} && (13)\\
&= (a_0^{d-1} a_1)^{d_1} \cdots (a_0^{d-1} a_n)^{d_n} && (14)\\
&= (a_0)^{(d-1)d_1} \cdots a_0^{(d-1)d_n} a_D && (15)\\
&= a_0^{(d-1)d} a_D && (16)\\
&= a_{(d,\ldots,0)}^{d-1} a_D && (17)\\
&= a_D && (18)
\end{aligned}$$

where (14) and (18) are a result of the fact that we set $a_0 = 1$. Therefore $b \in image(F_{d,n}(\mathbb{P}^n))$ and $X_{d,n} \subseteq F_{d,n}(\mathbb{P}^n)$. ℵ

Whats good is that we get explicit defining polynomials but we can even show that Veronese maps are in fact isomorphism of $X_{d,n}$ and $\mathbb{P}^n$.

Proposition 2.2. *For any $d, n \geq 1$, the Veronese map $F_{d,n} : \mathbb{P}^n \to X_{d,n}$ is an isomorphism.*

Proof. To prove that this is an isomorphism we construct the following map for any $b \in X_{d,n}$ define

$$G_{d,n}(b) = \begin{cases} [b_{(d,0,\ldots,0)} : b_{(d-1,1\ldots,0)} : \cdots : b_{(d,0,\ldots,1)}] \\ [b_{(1,d-1,\ldots,0)} : b_{(0,d,\ldots,0)} : \cdots : b_{(0,d-1,\ldots,1)}] \\ \qquad \vdots \\ [b_{(1,0,\ldots,d-1)} : b_{(0,1,\ldots,d-1)} : \cdots : b_{(d,0,\ldots,d)}] \end{cases}$$

where we choose the correct presentation depending on the point b. In the previous proof we saw that not every component could be 0 so we know that the point given by G is well defined. It remains to check the composition but that seems tedious and I am happy to accept that this is an inverse. ℵ

Ex: Let $X = \mathcal{V}(x_0^3 + x_1^2 + x_2^3) \subseteq \mathbb{P}^2$ and we consider the Veronese map

$$F_{3,2} : X \to \mathbb{P}^{\binom{5}{3}-1} = \mathbb{P}^9$$

Observe that $D = (d_0, d_1, d_2)$ is the tuple indexing outputs of the Veronese map and $d_1+d_2+d_3 = 3$. Then we can verify that

$$a \in X \iff a_0^3+a_1^3+a_2^3 = 0 \iff a_{(3,0,0)}+a_{(0,3,0)}+a_{(0,0,3)} = 0 \iff F_{3,2}(a) \in \mathcal{V}(y_{(3,0,0)}, y_{(0,3,0)}, y_{(0,0,3)})$$

then

$$F_{3,2}(X) \cong X_{3,2} \cap \mathcal{V}(y_{(3,0,0)}, y_{(0,3,0)}, y_{(0,0,3)})$$

following from the previous proposition. More generally we can present the first most important consequences of the relationship given by Veronese maps.

Proposition 2.3. *Up to isomorphism, every projective variety can be written as the vanishing set of a finite collection of homogenous polynomials of degree at most two.*

Proof. Let $X \subseteq \mathbb{P}^n$ be a projective variety. The primary observation is that X can be expressed as the vanishing set of polynomials $f_1, \dots, f_k$ of the same homogenous degree. Then each f_i has the form

$$f_i = \sum_D a_{i,D} x_0^{d_0} \cdots x_n^{d_n}$$

over all tuples $D = (d_0, \dots, d_n)$ *verify this claim*. Now consider the Veronese map

$$F_{d,n} : \mathbb{P}^n \to \mathbb{P}^{\binom{n+d}{d}-1}$$

and define linear polynomials $\ell_1, \dots, \ell_k$ such that

$$\ell_i = \sum_D a_{i,D} y_D$$

We are trying to get a Veronese map whose image is a variety which lets us treat the map as an ismorphism Then we observe that

$$a \in X \iff F_{d,n}(a) \in X_{d,n} \cap \mathcal{V}(\ell_1, \dots, \ell_k)$$

which implies that the image of the regular map on the variety X is exactly this intersection. Since the map is regular it can be restricted to a regular map on X. But because we know that the Veronese map is an isomorphism onto the variety $X_{d,n}$ then we have a regular invese to the map giving us an isomorphism $X \cong X_{d,n} \cap \mathcal{V}(\ell_1, \dots, \ell_k)$ where everthing defining this isomorphic variety is given by quadratics and linear polynomials. ℵ

This result is particularly interesting because we can say that up to isomorphism, polynomial maps between variety are linear.

Proposition 2.4. *If $F : X \to Y$ a polynomial map of projective varieties, then there exists an isomorphism $G : X \to Z$ and a linear map $L : Z \to Y$ such that $F = L \circ G$.*

Proof. *Exercises - in HW5* ℵ

With this and exercise 10.2.6 we can impose a rather strong condition on polynomial maps.

Proposition 2.5. *A $F : X \to Y$ is a polynomial map of projective varieties, then for any $b \in Y$, there are finitely many $a \in X$ such that $F(a) = b$.*

Proof. Using the previous proposition, $F = L \circ G$ for some linear map L and isomorphism G. Because isomorphisms are one-to-one and linear maps are finite-to-one then F must be finite-to-one. ℵ

2.4 10.4 Segre Maps

We have a solid understanding of projective varieties and ways in which we can embed them into higher dimensional spaces. Now what if have $\mathbb{P}^m \times \mathbb{P}^n$, with our affine intuition we wonder: Does this naturally live in $\mathbb{P}^{m+n}$? The way in which I interpret this is to think about a term $a \in \mathbb{P}^{m+n}$ where the first m terms are zero. If we were to break this back down into the product structure this would leave us with an all zero point in $\mathbb{P}^m$ which doesn't make sense. Fortunately for us, Segre has devised a way to embed this product into a higher projective space via a Segre map.

Definition 2.5 (Segre Map)**.** *For $m, n \geq 0$ the Segre map $S_{m,n} : \mathbb{P}^m \times \mathbb{P}^n \to \mathbb{P}^{(m+1)(n+1)-1}$ is a map*

$$S_{m,n} = [x_0y_0 : x_0y_1 : \cdots : x_my_{n-1} : x_my_n]$$

where the monomials are all possible products x_iy_j.

Observe that if all products $a_ib_j = 0$ take $b_j = 0$ then for all i, $a_ib_j = 0$ which means that a_i is not necessarily zero for all i. Because we have any other term a_ib_k and we take $a_i \neq 0$ then we force all b_j to be zero. So we only get zero if $a_i = 0$ for all i or $b_i = 0$ for all i. Because this can't happen in $\mathbb{P}^n$ or $\mathbb{P}^m$ then we are good and the Segre map is well defined. Lets look at a simple example

$$S_{1,1} : \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3 \tag{19}$$

$$([a_0 : a_1], [b_0 : b_1]) \mapsto [a_0b_0 : a_0b_1 : a_1b_0 : a_1b_1] \tag{20}$$

Observe that in general we can organize entries in $\mathbb{P}^{(m+1)(n+1)-1}$ into components z_{ij} where $0 \leq i \leq m$ and $0 \leq j \leq n$. These entries can be used to populate a matrix $A(z)$. Using our previous example we take $z = [a_0b_0 : a_0b_1 : a_1b_0 : a_1b_1]$

$$A(z) = \begin{bmatrix} a_0b_0 & a_0b_1 \\ a_1b_0 & a_1b_1 \end{bmatrix}$$

Using this presentation we can give the cartesian product of projective spaces by embedding them as the vanishing set in a higher dimensional projective space using a Segre map.

Proposition 2.6. *$S_{m,n}$ is an injection of $\mathbb{P}^m \times \mathbb{P}^n$ as a projective variety $Z_{m,n} \subseteq \mathbb{P}^{(m+1)(n+1)-1}$ where*

$$Z_{m,n} = \mathcal{V}(\{z_{ij}z_{k\ell} - z_{i\ell}z_{kj} : 0 \leq i, k \leq m, 0 \leq j, \ell \leq n\}).$$

Proof. We will begin by showing that $\operatorname{im}(S_{m,n}) = Z_{m,n}$. Take $(a, b) \in \mathbb{P}^m \times \mathbb{P}^n$ and take $z = S_{m,n}(a, b)$ then for $0 \leq i, k \leq m$ and $0 \leq j, \ell \leq n$ we see that

$$z_{ij}z_{k\ell} - z_{i\ell}z_{kj} = a_ib_ja_kb_\ell - a_ib_\ell a_kb_j = 0$$

hence $\operatorname{im} S_{m,n} \subseteq Z_{m,n}$. For $c \in Z_{m,n}$ implying that all 2×2 minors of $A(c)$ vanish. Then $\operatorname{rk} A(c) = 1$ because if the rank was zero this would imply that c was zero in every component. Since $\operatorname{rk} A(c) = 1$ then every row is a multiple of a single row. Let this single row be $b = (b_0, \ldots, b_m)$ and take every other row to be a_ib for some $a_i \in K$. Then we see that

$$c = S_{m,n}([1 : a_1 : \cdots : a_m], [b_0 : \cdots : b_n])$$

hence $\operatorname{im}(S_{m,n}) = Z_{m,n}$. Now we show that $S_{m,n}$ is injective. Consider $z, z' \in Z_{m,n}$ such that $z = S_{m,n}(a, b)$ and $z' = S_{m.n}(x, y)$ for some some $a, x \in \mathbb{P}^m$ and $b, y \in \mathbb{P}^n$ where $z = z'$. This implies that $A(z) = A(z')$, and as we saw in the previous part of this proof, the fact that each 2×2 minor vanishes in both matrices means that they must be rank 1. In particular this means that the first row of $A(z)$ can be set to b and each successive row set to a_ib. Similarly the first row of $A(z')$ can be set to y and the successive rows are multiples x_iy. Since these matrices are equivalent this means that $b = y$. Then $a_ib = x_iy = x_ib$ which implies that $x_i = a_i$ for all i. Therefore $(a, b) = (x, y)$ and we can conlcude that the Segre map is injective. ℵ

Now that we can view a product of the projective spaces as varieties, does this mean that taking products of the varieties of these spaces can be injected to a variety of a higher projective dimension.

Proposition 2.7. *If $X \subseteq \mathbb{P}^m$ and $Y \subseteq \mathbb{P}^n$ are projective vareities, then the Segre map $S_{m,n}$ resitrcts an injective variety $X \times Y$ onto a projective variety in $\mathbb{P}^{\binom{n+d}{d}-1}$.*

Proof. Let $X = \mathcal{V}(\mathcal{S})$ and $Y = \mathcal{V}(\mathcal{T})$ be projetive varieties for polynomials $\mathcal{S} \subseteq K[x_0, \ldots, x_m]$ and $\mathcal{T} \subseteq K[x_0, \ldots, x_n]$. For all $f \in \mathcal{S}$ and $g \in \mathcal{T}$ define

$$f_i(z) = f(z_{0i}, \ldots, z_{mi}) \quad \text{and} \quad g_j(z_{j0}, \ldots, z_{jn})$$

collecting all $f_i \in \mathcal{S}'$ and $g_j \in \mathcal{T}'$. Then for $(x, y) \in X \times Y$ we know that $S_{m,n}(x, y) \in Z_{m,n}$ and observe that for all i

$$f(x_0 y_i, \ldots, x_m y_i) = y_i^d f(x_0, \ldots, x_m) = 0$$

and

$$g(x_j y_0, \ldots, x_j y_n) = x_j^d g(y_0, \ldots, y_m)$$

then $S_{m,n}(X \times Y) \subseteq Z_{m,n} \cap \mathcal{V}(\mathcal{S}' \cup \mathcal{T}')$. Reversing this process we can check again that vanishing on $\mathcal{S}', \mathcal{T}'$ will imply that the original points vanish on $\mathcal{S}$ and $\mathcal{T}$. ℵ

With this we can define the product of projective varieties.

Definition 2.6. *If $X \subseteq \mathbb{P}^m$ and $Y \subseteq \mathbb{P}^n$ are projective varieties then their product $X \times Y$ is the projective variety*

$$S_{m,n}(X \times Y) \subseteq \mathbb{P}^{(m+1)(n+1)-1}$$

Now we ask whether maps into or out of products of varieities are regular. In particular, our soon to be topological perspecitve, wants us to determine whether the projection maps of $X \times Y$ are regular.

Proposition 2.8. *If $X \subseteq \mathbb{P}^m$ and $Y \subseteq \mathbb{P}^n$ are projective varieites. Then the projection maps $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ are regular.*

Under the Segre map we may note that for z we can always find m terms $z_{i,0}, \ldots, z_{i,m}$ which when mapped into Y gives us a point scaled by x_j which is equivalent to a point in X.

Proof. We construct π_1 and π_2 follows immdietly from symmetry. For $c = (a, b) \in X \times Y \subseteq \mathbb{P}^{(m+1)(n+1)-1}$ define

$$\pi_1(c) = \begin{cases} [c_{00} : c_{10} : \cdots : c_{m0}] & c \notin \mathcal{V}(z_{00}, \ldots, z_{m0}) \\ \vdots & \\ [c_{0n} : c_{1n} : \cdots : c_{mn}] & c \notin \mathcal{V}(z_{0n}, \ldots, z_{mn}) \end{cases}$$

For $c = (a, b) \in X \times Y$, at least one a_i or b_j is non-zero. Without loss of generality take $b_j \neq 0$ then $c \notin \mathcal{V}(z_{0j}, \ldots, z_{nj})$ and

$$\pi_1(c) = [a_0 b_j : \cdots : a_n b_j] = [a_0 : \cdots : a_n].$$

ℵ